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CONTEXT 4
MAKING REACTIONS WORK FOR US

By the end of this chapter you will have covered the following material.

Science as a Human Endeavour

- Development of complex models and/or theories often requires a wide range of evidence from multiple individuals across disciplines (ACSCH010 and ACSCH049)
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)
- The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)
- Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053)

Contextual story – Making reactions work for us

- The rates of reactions can be altered.
- The rates of reactions in our bodies depend on concentration, temperature, surface area and catalysts.
- We can control the rates of reactions in and around our home and in industry.

Chemical understanding

- Reaction rates
- The collision theory
- The factors that affect the rate of a reaction:
  - Concentration
  - Pressure
  - Temperature
  - Surface area
  - Catalyst
- Kinetic theory of gases
- Calculations involving gases

Investigations

Context 4

- Activities 4.1, 4.2, 4.3, 4.4, 4.5 and 4.6
- Experiments 4.1, 4.2 and 4.3
- Investigation 4.1

Chapter 9

- Activities 9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.7, 9.8, 9.9 and 9.10
- Experiments 9.1, 9.2, 9.3, 9.4, 9.5 and 9.6
- Investigation 9.1

Science as a Human Endeavour (SHE)

- Development of complex models and/or theories often requires a wide range of evidence from multiple individuals and across disciplines (ACSCH010 and ACSCH049)
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)
- The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)
- Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053)

Science Understanding (SU)

- The behaviour of gases, including the qualitative relationships between pressure, temperature and volume, can be explained using kinetic theory (ACSCH060)
- Varying the conditions present during chemical reactions can affect the rate of the reaction and in some cases the identity of the products (ACSCH068)
- The rate of chemical reactions can be quantified by measuring the rate of formation of products or the depletion of reactants (ACSCH069)
- Collision theory can be used to explain and predict the effect of concentration, temperature, pressure and surface area on the rate of chemical reactions by considering the structure of the reactants and the energy of particles (ACSCH070)
- The activation energy is the minimum energy required for a chemical reaction to occur and is related to the strength of the existing chemical bonds; the magnitude of the activation energy influences the rate of a chemical reaction (ACSCH071)
- Energy profile diagrams can be used to represent the enthalpy changes and activation energy associated with a chemical reaction (ACSCH072)
- Catalysts, including enzymes and metal nanoparticles, affect the rate of certain reactions by providing an alternative reaction pathway with a reduced activation energy, hence increasing the proportion of collisions that lead to a chemical change (ACSCH073)

Science Inquiry Skills (SIS)

- Identify, research, construct and refine questions for investigation; propose hypotheses; and predict possible outcomes (ACSCH040)
- Design investigations, including the procedure/s to be followed, the materials required, and the type and amount of primary and/or secondary data to be collected; conduct risk assessments; and consider research ethics (ACSCH002 and ACSCH041)
- Conduct investigations, including measuring pH and the rate of formation of products, identifying the products of reactions and testing solubilities, safely, competently and methodically for the collection of valid and reliable data (ACSCH042)
- Represent data in meaningful and useful ways, including using appropriate graphic representations and correct units and symbols; organise and process data to identify trends, patterns and relationships; identify sources of random and systematic error, identify anamolous data and estimate their effect on measurement results; and select, synthesise and use evidence to make and justify conclusions (ACSCH043)
- Select, construct and use appropriate representations, including physical and graphical models of molecules, energy profile diagrams, electron dot diagrams, ionic formulae, chemical formulae, chemical equations and phase descriptors for chemical species to communicate conceptual understanding, solve problems and make predictions (ACSCH045)
- Communicate to specific audiences and for specific purposes using language, nomenclature, genres and modes, including scientific reports (ACSCH008 and ACSCH047)

Reactions

Imagine if you were in a car crash and the air bag inflated too slowly. Or if the metal frame of a building rusted as soon as it was erected. Or if the food we eat took weeks to digest. Many reactions affect our everyday lives; however, it is not only what reactions happen, but how quickly they happen that is important. As our understanding of chemical reactions has increased, we have been able to understand why reactions occur at the rate that they do. This has allowed chemists to manipulate reactions so they occur at the rate that we need.

▲ **Figure 4.1**
Rust has destroyed this building. Rusting is a slow reaction.

Reactions in our bodies

The human body is an amazing system that relies on many chemical reactions to make it work efficiently. These reactions need to occur at the correct **rate** to keep us healthy. Otherwise, many diseases can occur.

WOW

Maple syrup urine?

One in 180 000 people lack the enzyme that breaks down branched-chain amino acids, including leucine, isoleucine and valine. This leads to a build-up of these amino acids and their toxic by-products. People with this genetic disease produce a sweet-smelling urine, similar to the smell of maple syrup, hence the name Maple Syrup Urine disease. Affected individuals need to eat a modified diet that is reduced in these amino acids to prevent loss of appetite, vomiting, lethargy, seizures, coma and possible death.

To learn about how reaction rates are measured, refer to Chemistry section 9.1 on page 368.

To learn about collision theory, refer to Chemistry section 9.2 on page 374.

COLLISION THEORY

View this animation and change the conditions to see what happens in a collision where there is not enough energy, the correct energy and orientations, and the correct energy but the wrong orientation.

Biochemistry

The study of chemical processes in living organisms is known as **biochemistry**. This is a large area combining chemistry and biology. It involves the study of chemicals in plants and animals and the reactions that they are involved in. This ever-expanding field of science links genetics, metabolism, hormones, transport across cell membranes, immunity and many diseases.

Biochemistry includes all aspects of the chemical reaction – the reactants, the products, the process and the rate. All these factors are important in understanding the way our bodies work. This chapter focuses on the rate of the reactions; however, rate is related to these other factors.

To understand why some reactions occur quickly, and others slowly, we must understand what happens during a chemical reaction. **Collision theory** is a model that allows us to explain what happens during a chemical reaction and the factors that affect it.

Enzymes

Chemical reactions occur every day in your body to keep it healthy. Many of these reactions rely on **enzymes** to allow the reaction to occur at an adequate rate. An enzyme is a biological catalyst. This means that it is found in living organisms and, being a catalyst, can speed up a chemical reaction without being consumed.

For more information about catalysts, refer to Chemistry section 9.3 on page 379.

Enzymes are usually large protein molecules. Enzymes are highly specific and act on a particular reaction, type of reaction or bond type by making it easier for the reaction to occur. The enzyme has an **active site** with a shape that matches the reactant particles. When the enzyme and reactants (known as the **substrates**) bind, a change occurs which allows the reactants to form the products in a way that lowers the activation energy. Because less energy is needed for a successful collision, more particles have enough energy and therefore more collisions will result in a chemical reaction.

Our digestive system provides the nutrients we need for growth, repair and energy. The food we eat is largely made up of **carbohydrates**, **proteins** and **fats**. These are large organic molecules that are made up of smaller units (Table 4.1). The digestive system breaks down the large molecules into smaller molecules that can be absorbed into the bloodstream. In the blood, the molecules are transported to where they are needed. However, if the large molecules cannot be broken down, they cannot be absorbed and therefore the food cannot be used by the body.

HOW ENZYMES WORK

View this animation of models of various ways enzymes work.

Table 4.1 Nutrients in food

Nutrient	Small units that can be absorbed
Proteins	Amino acids
Carbohydrates	Simple sugars; for example, glucose
Fats	Glycerol and fatty acids

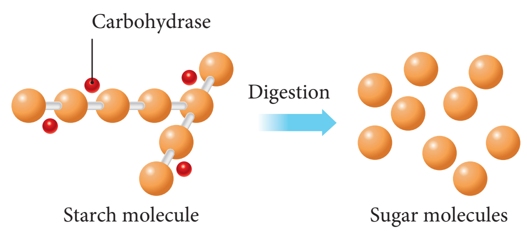


Figure 4.2 Enzymes help break complex molecules into smaller, simpler molecules that can be absorbed. Carbohydrase acts on starch molecules to produce sugar molecules.

Enzymes can be specific for a particular reaction, breaking a particular bond in a particular molecule (Figure 4.3). This means that the larger molecules are split into smaller sections. Upon binding to the enzyme at the active site, the substrate is changed in such a way as to allow the bond to be broken with less energy. The products formed are then released and the enzyme is available for further reactions. Table 4.2 on page 87 lists some features of some digestive enzymes.

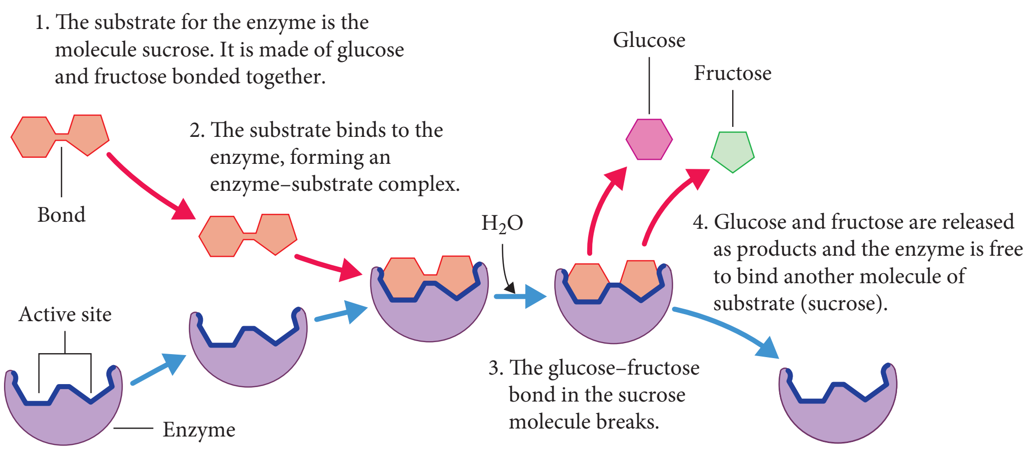


Figure 4.3 A model of an enzyme

You will learn more about proteins, carbohydrates and fats in Units 3 & 4.

Table 4.2 Digestive enzymes

Enzyme	Location	Reactant	Product
Amylase	Saliva in the mouth Pancreas releases it into the small intestine	Starch (a large carbohydrate)	Maltose
Protease (pepsin)	Stomach	Proteins	Smaller proteins
Protease (trypsin)	Pancreas releases it into the small intestine	Proteins	Peptides and amino acids
Lipase	Pancreas releases it into the small intestine	Fats	Fatty acids and glycerol
Peptidase	Small intestine	Peptides (small sections of protein)	Amino acids
Sucrase	Small intestine	Sucrose (a small carbohydrate)	Glucose and fructose
Maltase	Small intestine	Maltose (a small carbohydrate)	Glucose
Lactase	Small intestine	Lactose (a small carbohydrate)	Glucose and galactose

There are thousands of different enzymes in the human body controlling the wide range of reactions that are occurring. These include:

- carbonic anhydrase, which catalyses the reaction between carbon dioxide and water to form carbonic acid, allowing carbon dioxide to travel through the blood: $CO_2(g) + H_2O(l) \rightarrow H_2CO_3(aq)$
- DNA polymerase, which is involved in DNA replication
- dehydrogenase, an important enzyme in the first step of cellular respiration in mitochondria to produce energy in our bodies
- cyclin dependent kinases, which control cell division
- tyrosinase, which triggers melanin synthesis and is important in determining skin colour.

The activity or effectiveness of enzymes can be affected by the conditions or environment in which they are found. These conditions include:

- concentration – the higher the enzyme concentration, the more substrate that can bind to an enzyme at a given time and the greater the reaction rate
- temperature – enzymes are proteins, and as such are **denatured** above about 45°C. Above this temperature, the structure of the protein changes, changing the shape of the active site. This permanently inactivates the enzyme as the substrate is unable to bind to the active site
- pH – different enzymes require different pH values for optimum activity; for example, the pH of the stomach is much less than the pH of the rest of the digestive system, so only specific enzymes will be activated in the stomach
- **co-enzymes** – ions or non-protein molecules such as vitamins that change the shape of the active site in order for the enzyme to function.

WOW

Coenzyme Q10

Coenzyme Q10 is a naturally occurring enzyme that is found in all cells. However, it is more abundant in the hardest working cells such as heart muscle cells. Its functions include acting as an antioxidant and improving the efficiency of energy production. Research is being done into the effects of taking coenzyme Q10 supplements on conditions such as heart disease, cancer and Parkinson's disease.

EXPERIMENT 4.1

EFFECT OF pH AND TEMPERATURE ON THE ENZYME COMPLEX RENNET

Rennet is a complex of enzymes naturally found in the stomachs of mammals. It helps the coagulation of milk – separating it into a solid and a liquid component. It is especially important in the digestion of milk in young animals.

Rennet is also utilised in milk desserts in the form of a setting agent known as junket. The enzymes help the milk set, forming a soft solid.

Aim

To investigate the effect of pH and temperature on the effectiveness of the enzymes in rennet

Materials

- 1 packet of junket (This may vary depending on the quantity that can be made – you will need enough for 65 mL of milk)
- 100 mL of milk
- Distilled water
- 1 mL of 1 mol L^{-1} hydrochloric acid (HCl)
- 1 mL of 1 mol L^{-1} acetic acid (CH_3COOH)
- 1 mL of 1 mol L^{-1} sodium acetate ($NaCH_3COO$)
- 1 mL of 1 mol L^{-1} sodium hydroxide (NaOH)
- 13 test tubes
- Test tube rack
- Water baths at 10°C, 20°C, 30°C, 40°C, 50°C, 60°C, 70°C and 80°C
- 10 mL measuring cylinder
- Thermometer
- Stopwatch
- pH meter

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.
1 mol L^{-1} HCl is corrosive to skin and clothes.	Wear gloves and aprons.
1 mol L^{-1} NaOH is corrosive to skin and clothes.	Wear gloves and aprons.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

Part A: Effect of temperature

1. Place 5 mL of milk into each of eight test tubes.
2. Place one test tube in each water bath and leave it until the milk has reached the desired temperature.
3. Dissolve the junket in 15 mL of water. (You will need to keep some of this mixture for part B.)
4. Place 1 mL of junket into the first test tube, mix and record the temperature inside the test tube, and start the stopwatch immediately.
5. Stop the stopwatch when the milk has become firm. Record the time taken to set.
6. Repeat steps 4 and 5 for each of the milks at the other temperatures.

Part B: Effect of pH

7. Place 5 mL of milk into each of four test tubes.
8. Record the temperature of the milk – these should all be the same.
9. Into test tube 1, place 1 mL of distilled water.
10. Into test tube 2, place 1 mL of 1 mol L^{-1} HCl.
11. Into test tube 3, place 1 mL of 1 mol L^{-1} CH_3COOH.

12 Into test tube 4, place 1 mL of 1 mol L^{-1} NaOH.

13 Into test tube 5, place 1 mL of 1 mol L^{-1} $NaCH_3COO$.

14 Use the pH meter to test the pH of the milk in each test tube. Ensure that the meters are cleaned thoroughly between test tubes. Record the pH.

15 Add 1 mL of the junket solution from part A to test tube 1 and time how long it takes the milk to become firm. Record the time taken.

16 Repeat step 15 for test tubes 2–5.

Results

Construct a results table like this in your notebook. Record your data in the table.

Variable	Test tube	Temperature (°C)	pH	Time for milk to become firm (min)
Temperature	1			
	2			
	3			
	4			
	5			
	6			
	7			
	8			
pH	1 (distilled water)			
	2 (HCl)			
	3 (CH_3COOH)			
	4 (NaOH)			
	5 ($NaCH_3COO$)			

Analysis of results

1 Draw separate graphs for parts A and B.

2 What happened to the time taken for milk to become firm as the temperature increased?

3 What happened to the time taken for the milk to become firm as the pH increased?

4 Do these results support your understanding of the effect of temperature and pH on enzymes?

Discussion

1 What aspects of the experiment increased the accuracy and reliability of the results?

2 What were the major difficulties in obtaining accurate results during the experiment?

3 How could the experiment be improved?

Conclusion

1 What conclusion can you make about the effect of temperature on the activity of enzymes in junket?

2 What conclusion can you make about the effect of pH on the activity of the enzymes in junket?

Taking it further

1 In what other ways could you investigate the action of enzymes?

2 Research the role of rennet (or rennin). Explain how your findings about the effect of temperature and pH may play a role in ensuring the enzyme's effectiveness.

Figure 4.4 ▲ Pineapples contain bromelain – an enzyme that is proteolytic (breaks down proteins).

Enzymes in our food

Many of our foods come from plants and animals, which contain enzymes. These enzymes are still present in the food we eat.

Pineapples contain the enzyme bromelain. This is a **proteolytic enzyme**, meaning it breaks down proteins. Proteins are made up of many **amino acids** joined together into a very large molecule. Although there are only 20 amino acids, they can combine in many different combinations to produce many different proteins. Proteolytic enzymes, also known as proteases, are able to break the bonds between the amino acids, thereby breaking up the protein. The breakdown of proteins can occur without the presence of proteases, but at a much slower rate.

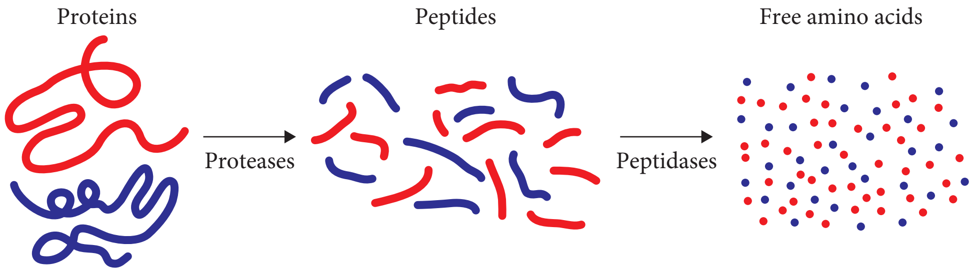


Figure 4.5 ▶ Proteins are broken down into smaller sections called peptides, which are further broken down into individual amino acids.

EXPERIMENT 4.2

ENZYMES IN PINEAPPLE

Pineapples contain the enzyme bromelain, which breaks down the bonds between the amino acids in proteins.

Jelly contains the protein gelatin, which makes jelly set. The enzymes in pineapple are able to break this protein down, causing the jelly to become a liquid.

Aim

To observe the action of the enzyme bromelain in pineapples

Materials

- Fresh pineapple, sliced and cut into quarters
- Jelly crystals
- 2 × 250 mL beakers
- Spatula
- Hot and cold water

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

1. Follow the instructions on the jelly packet to make the jelly.
2. Divide the jelly between the two 250 mL beakers.
3. Refrigerate the jelly until it is nearly, but not completely, set.

4 Place a piece of fresh pineapple on the top of the jelly in one of the beakers.
5 Refrigerate the beakers overnight.
6 Record your observations of the jelly in each of the beakers.

Results

Record your results in an appropriate table. Include an appropriate title for your table.

Analysis of results

1 In which beaker did the jelly remain firm?
2 In which beaker did the jelly turn into a liquid?
3 Use collision theory to provide an explanation for how the enzyme in fresh pineapple is able to speed up the breakdown of proteins in the jelly.

Discussion

In what ways could the experiment be improved to improve the accuracy and reliability of the data collected?

Conclusion

What conclusion can you make about the activity of the enzymes in pineapple?

Taking it further

How could you test the effect of different treatments on the activity of the enzymes in pineapple; for example, temperature and pH?

ACTIVITY 4.1

TENDERISING STEAK

What makes some steaks tender, and other steaks tough?

What you will need

- Computer with Internet access

What to do

1 Read the first two pages of the article located at the weblink 'Tender steak'.
2 Explain what it is that makes meat tough, and how it is made tender.
3 List the different methods of tenderising steak that are mentioned in the article, and explain how each method works.
4 Use your knowledge of digestive enzymes to explain how the steaks are tenderised by enzymes.
5 Use collision theory to explain why meat can be tenderised quickly using pineapple, but it takes days if it is just left out of the fridge.

TENDER STEAK

Read the article to find out what makes steak tough and how to make it tender.

Energy production

Our bodies require energy for just about everything – for our brains to work, for our muscles to maintain posture, for our heart to pump, for movement and to maintain our body temperature.

Cellular respiration is a chemical reaction occurring in cells that produces energy from the food we eat and oxygen we breathe. More specifically, the reactants are glucose and oxygen while the products are carbon dioxide, water and energy.

Word equation: Glucose + oxygen → carbon dioxide + water + energy

Chemical equation: $C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l) + \text{energy}$

To learn about what affects rate of reaction, refer to Chemistry section 9.3 on page 379.

It is important that this reaction occurs at a rate that matches the needs of the body. The body can ensure this happens by controlling the concentration of the reactants – oxygen and glucose. The higher the concentration of the reactants, the greater the chance that the reactant particles will collide. A greater number of collisions mean a greater number of successful collisions and therefore reaction rate increases.

After skipping a meal, you often feel tired and lethargic. This is due to a reduced blood glucose concentration, lowering the rate of cellular metabolism and therefore energy production. People who climb mountains often feel very light-headed, nauseous, tired and lethargic and have trouble thinking clearly. This is due to the reduced oxygen concentration in the air at higher altitudes. This follows through to a reduction in the blood oxygen concentration, rate of cellular metabolism and therefore energy production.

To learn about gas pressure and amount of gases in the atmosphere, refer to Chemistry section 8.1 on page 342.

WOW

Altitude sickness

Atmospheric pressure decreases the higher up a mountain you travel. This means that there are fewer oxygen molecules available – at 3600 m above sea level there are only 60% of the number of oxygen molecules per breath. This reduction in oxygen along with the lower atmospheric pressure can lead to altitude sickness if you don't acclimatise your body as you travel up the mountain.

There are three types of altitude sickness – acute mountain sickness, high-altitude pulmonary oedema and high-altitude cerebral oedema. The symptoms range from headaches, nausea, shortness of breath to decreased coordination, tightness of the chest, marked fatigue, confusion, decreased consciousness and coma. Many of these symptoms are a direct result of a lack of oxygen and the body trying to increase its oxygen intake.

To learn about the effect of concentration on reaction rates, refer to Chemistry section 9.3 on page 379.

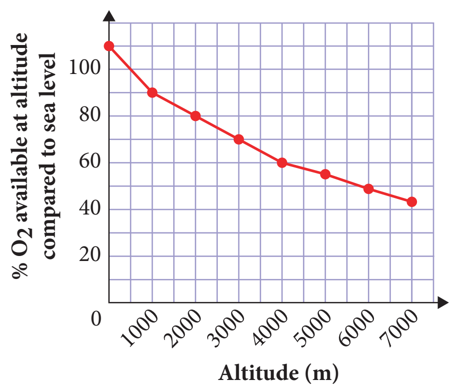


Figure 4.6 Oxygen availability at different altitudes

Homeostasis involves complex processes that control our internal environment, including oxygen and glucose concentrations. The rate of breathing controls the amount of oxygen entering our blood and which is transported to cells. Blood glucose concentration is controlled by the hormones insulin and glucagon from the pancreas. These concentrations are also influenced by chemical equilibrium, which will be considered in Units 3 & 4.

WOW

Homeostasis

Homeostasis is an important area of study in Biology and Human Biology. It considers the various mechanisms that are involved in maintaining a constant internal environment. This is to keep factors such as temperature, pH, gas concentrations, blood glucose and water within the optimum limits so that our bodies function correctly. In many instances, this is directly linked to chemical reactions being able to occur at the required rates.

Why we chew our food

◀ **Figure 4.7** By cutting the food into smaller pieces, a greater surface area is available for reaction.

iStockphoto/jsemeniuk

iStockphoto/Forzua

In most cases, foods contain large, complex molecules. For your body to use these, they must be broken down into small, simple molecules that can be absorbed and then used for the different processes in the body. A lot has to happen to food before your body can use it.

The process of digestion starts in the mouth. Food must be chewed thoroughly to make it easy to swallow. The second reason that you need to chew food is to speed up its digestion. By breaking up the food into smaller pieces, chewing increases the surface area of the food that is exposed to the digestive chemicals. This means that more of the food can be digested at the same time, decreasing the time it takes for the meal to be digested overall.

Chemical digestion can only occur when the digestive juices meet the food particles. These particles must collide so that the reaction can take place. In a solid, this can only occur on the surface because the interior particles are not able to collide with the digestive juices. When the food is broken up into smaller pieces, the particles on the inside become exposed, and are able to react.

EXPERIMENT 4.3

WHY WE CHEW OUR FOOD

Chewing food breaks up large pieces into smaller pieces. This is known as physical digestion and has the function of increasing the surface area of the food. This allows more of the food to be exposed to digestive enzymes and so will speed up chemical digestion.

In this experiment, agar, with phenolphthalein, will represent the food. The phenolphthalein is an indicator that will initially be pink (neutral pH). In an acidic solution, this indicator will become colourless. This will allow you to measure how much of the 'food' has undergone chemical digestion.

Aim

To investigate the effect of surface area on the rate of digestion

Materials

- Agar, with phenolphthalein, cubes: 16 × 1 cm sides, 4 × 2 cm sides, 1 × 4 cm sides
- Vinegar (volume will depend on size of containers)
- 3 containers deep enough for cubes to be submerged in liquid
- Ruler
- Tongs
- Paper towel

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Chemicals could splash in your eyes.	Wear safety glasses at all times.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

1 Place all the 1 cm cubes into the first container, ensuring that they are not touching each other.
2 Pour vinegar over the cubes until they are covered.
3 Leave the cubes in the acid for 20 minutes.
4 Use tongs to carefully remove the cubes from the acid and use the paper towel to pat them dry.
5 Cut each cube in half and measure how far from the edge the colour changes from colourless to pink.
6 Repeat steps 1–5 for the other sized cubes.

Results

Record your results in an appropriate table. Remember to include an informative title for the table and appropriate headings on your columns.

Analysis of results

1 Calculate the percentage of each cube that has been 'digested'.
2 Calculate the total surface area for each size of cube. (Remember that surface area is the total area of all the faces of the cube.)
3 Graph your results of total surface area and the percentage digested in an appropriate graph.
4 What is the relationship between the surface area and the percentage digested?
5 Do these results support your understanding of the effect of surface area on the rate of a reaction?
6 Were there any anomalies in your results? If there were, what possible explanations are there for these anomalies?

Discussion

1 Why were there different numbers of cubes for the different sizes?
2 How does chewing (or cutting the agar up) increase the surface area?
3 What errors did you encounter during the experiment?
4 How could the experiment be improved to make it more accurate or reliable?

Conclusion

What conclusion can you make about the effect of surface area on the rate of digestion of food?

Taking it further

How could you further investigate the effect of surface area on the rate of reactions?

To learn about the effect of surface area on the rate of reactions, refer to Chemistry section 9.3 on page 379.

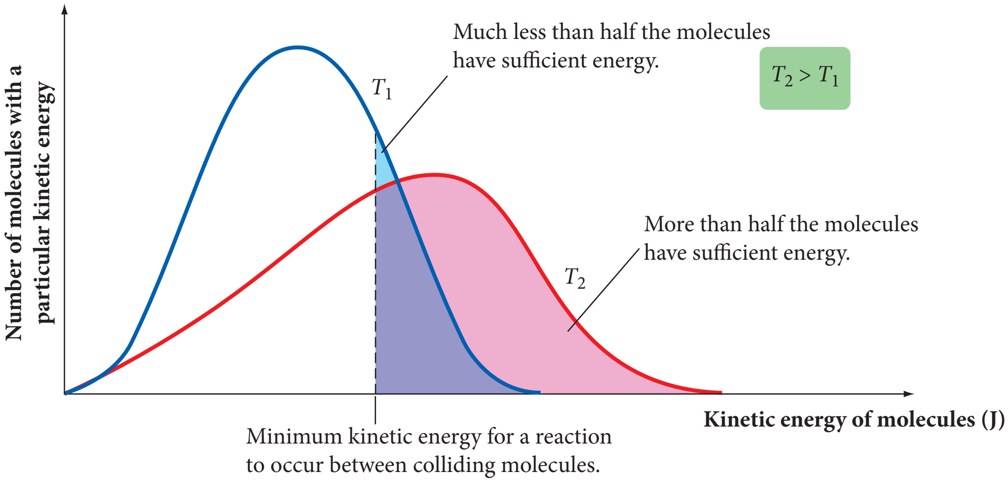
Another benefit of chewing is that it mixes the food with saliva, which starts the process of chemical digestion. Saliva contains enzymes, including amylase and lipase. Amylase is an enzyme for the breakdown of starch, a large carbohydrate molecule, into smaller carbohydrate molecules. Lipase, also an enzyme, is responsible for the breakdown of fats into smaller molecules. As with all enzymes, amylase and lipase allow these reactions to occur at a rate fast enough to be effective.

Hypothermia

Normally, the human body maintains a core temperature of about 37°C. However, in some situations the temperature may drop. **Hypothermia** occurs if the core body temperature drops below 35°C. The first sign of hypothermia is uncontrolled shivering as the body tries to produce heat. Other signs include drowsiness, loss of coordination, lethargy and difficulty breathing. If the core body temperature falls below 32°C, the person may slip into a coma. As the temperature continues to fall, the heart and lungs stop working effectively. This means that the cells do not get enough oxygen, and death may occur.

When the temperature falls, the rate of the reactions occurring in the body also decreases. This is because temperature is a measure of average **kinetic energy**. When the temperature decreases, the number of particles with enough energy for a successful collision decreases.

With fewer successful collisions, the reaction rate decreases. Kinetic energy is related to the velocity, or speed, of the particles. The particles are moving slower, so they will collide less frequently. This also decreases the number of successful collisions and therefore the reaction rate declines.



▲ **Figure 4.8**
The number of molecules with enough energy for a successful collision at different temperatures (*T*)

Hypothermia decreases the rate of the reactions occurring in the body, and therefore the body is unable to work efficiently. This includes cellular respiration, so the cells are not able to produce the energy needed to function properly.

To learn about the effect of temperature on the rate of reactions, refer to Chemistry section 9.3 on page 379.

To learn about the relationship between kinetic energy and temperature of gases, refer to Chemistry section 8.2 on page 346.

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052) AC

Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053) AC

Reactions around the home

There are many chemical reactions that we depend on to be able to carry out our everyday lives in our homes. Being able to control the rate of these reactions is vital for the efficiency of these reactions as well as our safety.

ACTIVITY 4.2

REACTIONS WHERE WE LIVE

What to do

1 Look around the inside and outside of your home.
2 List all the chemical reactions that take place that you can think of. Remember, a chemical reaction involves the production of a new substance.
3 For each reaction, identify the speed that we want the reaction to occur at (fast, moderate or slow).
4 Try to identify how we manipulate the reaction for it to occur at this desired rate.
5 Share your findings with others in your class.

What did you discover?

1 How reliant are we on chemical reactions in our everyday lives?
2 How are we able to control these reactions to our advantage?

123RF/Moodboard

Figure 4.9 ▲
Relaxing by a camp fire – but it's really chemistry!

Making fires

Before we had reverse-cycle air-conditioners, wood fires were often used to heat our homes. Today, they are still used in some locations for household heating, but are also popular for camping. These fires use the **combustion** reaction of wood, which produces heat and light due to the energy being released.

It is not always easy to get a fire started. To build a fire, you start with kindling, small pieces of wood that are usually ignited by burning some paper. Then, you add other pieces of wood of gradually increasing thickness. Once the fire is burning well, larger logs of wood are added so that they will continue burning slowly. This procedure is not random. It is all related to the rate of the reaction determined by the surface area available for the reaction.

ACTIVITY 4.3

FIRE AS A REACTION

What to do

1 Work through the interactive on the weblink 'On fire' to gain an understanding about what is happening in a fire.
2 a What are the reactants for a wood fire?
  b Why is kindling used to start a fire?
  c Why would large logs be used to keep a fire burning overnight?
  d Hospitals often have cylinders of oxygen gas. Why would it be very important to stop anyone smoking near the cylinders?
3 Go to the weblink 'How fire extinguishers work'. Read the four pages about how fire extinguishers work.
4 Use what you have learnt about fire extinguishers and your knowledge about collision theory and reaction rates to explain how a fire extinguisher can stop a fire.

What did you discover?

1 How is fire an example of a chemical reaction?
2 How can the rate of the reaction be manipulated in a fire?
3 How can our knowledge of reaction rates help us to extinguish fires?

ON FIRE
Learn more about combustion by exploring this virtual fire lab.

HOW FIRE EXTINGUISHERS WORK
Visit this website and read the article about fire extinguishers.

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and AACSCH052)

Preserving food

Have you ever taken the milk out of the fridge to make a milkshake, only to find that it has gone off? Or left a sandwich in your bag, only to find it weeks later by the smell? There are a variety of reasons that food will spoil. Methods have been developed to slow this degradation so that our food lasts longer.

Food spoilage happens when there is a change in the smell, look, taste or feel of the food that makes it inedible. There are two main reasons that food spoils: enzymes catalyse the breakdown of the food molecules; and **micro-organisms**, generally bacteria or fungi, cause the food to break down. In either case, if we can slow the breakdown of the food, we can make the food last longer.

Some **food preservation** methods are:

- freezing – keeping the food at temperatures below 0°C
- refrigeration – keeping the food at reduced temperatures, usually about 4°C
- canning – processing the food and then sealing it in an airtight container
- pickling – storing or cooking the food in a solution such as brine or vinegar that inhibits the growth of micro-organisms
- vacuum-packing – storing the food in a vacuum container.

Each of these methods utilises our understanding of chemical reactions and the conditions required by micro-organisms. By changing the conditions, we are able to slow the rate of the degradation reaction or microbial growth. This will mean that the food will take longer to spoil.

Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053) AC

ACTIVITY 4.4

FOOD PRESERVATION

Aim

To apply an understanding of reaction rates to understand how various food preservation methods work

You will need

- Computer with Internet access

What to do

1. Research what happens when food spoils. You may wish to use the information at the weblink 'Food spoilage' as a starting point.
2. Identify what the reactants are in the degradation of food. Be general; you do not need to be too specific.
3. Research each of the common methods of food preservation, finding out what is done and how it works. You may wish to use the information in the weblinks 'Methods of food preservation' and 'Food preservation' as a starting point.
4. Based on your research, identify which methods work by reducing microbial growth and which methods work by slowing the chemical reactions. Some methods may come under both classifications.
5. For each method that works by slowing the chemical reaction, identify which factor affects reaction rates.
6. Use your knowledge of collision theory to explain how each of the methods that affect the chemical reaction rate work.

What did you discover?

Explain how our understanding of the rates of reactions is able to help us to preserve food.

FOOD SPOILAGE

Visit this website to look at food spoilage in more detail.

METHODS OF FOOD PRESERVATION

Visit this website to learn more about commercial methods of preserving food.

FOOD PRESERVATION

Visit this website to learn more about preserving food.

WOW

Food poisoning

When appropriate food handling and preservation methods are not used, food may become contaminated with harmful micro-organisms. These include bacteria such as *Salmonella*, *Escherichia coli* and *Listeria*, viruses and toxins, which can cause food poisoning. Symptoms of food poisoning include nausea, vomiting, diarrhoea, stomach cramps, fever and headaches. In severe cases, it can cause long-term problems or even death. Each year, 5.4 million people in Australia suffer from food poisoning – something very dangerous that could be prevented!

INVESTIGATION 4.1

INVESTIGATING FOOD PRESERVATION

Many different methods are used to make food last longer. But which one is the most effective? Or how should the method be implemented to be the most effective? Your task is to consider an aspect of food preservation and design and conduct an investigation to further develop your understanding of food preservation and its relation to reaction rates.

What is your aim?

You may choose to compare methods of food preservation or choose one method and compare variations within that method. Your aim should be related to the effectiveness of food preservation of the different methods or variations.

What will you need?

Carefully consider what you will need to conduct a fair and thorough test. You also need to consider what you will need to maintain safety throughout your investigation.

What are the risks?

Construct a table similar to the one below. Identify specific risks involved in the investigation and ways that you will manage the risks to avoid injuries or damage to equipment. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?

How will you carry out your investigation?

Develop a detailed procedure for your investigation. Consider how you will change your independent variable and how you will measure your dependent variable. Ensure that all the other variables are controlled to make your investigation valid. You may wish to carry out a practice test to ensure that your method is going to be effective. Remember that a good method should detail exactly what you did to obtain your results.

What results will you collect?

Consider what data you will need to collect, how many trials you will need to conduct and the number of variations in your independent variable you will need.

How will you analyse your results?

What will you need to do with your data to form a conclusion?

What have you found?

What did your results show? Can you explain this by using your knowledge of food preservation? Are there any anomalies in your data? What are some possible explanations for these anomalies?

What can you conclude?

What conclusion can you make from your investigation?

Ideas for improvement

How could you improve your investigation to allow you to achieve your aim more accurately or thoroughly?

Taking it further

In what ways could you further investigate food preservation methods?

Batteries going flat

A battery uses a chemical reaction to produce electricity. There are many different batteries but, in all types, electrons are transferred from one reactant to another. This movement of electrons allows the production of electricity.

When a battery 'goes flat', the rate of the chemical reaction is too slow to be able to produce enough electricity.

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052) AC

The reaction that occurs in batteries is called a redox reaction. You will learn more about redox reactions in Units 3 & 4.

ACTIVITY 4.5

REACTIONS IN BATTERIES

Aim

To understand that batteries utilise a chemical reaction and that the rate of the reaction is related to batteries going flat

You will need

- Computer with Internet access

What to do

1. Choose one type of battery: zinc–carbon, alkaline, lithium ion or lead acid.
2. Conduct some research to find out the reactants that are involved in the reaction utilised in your chosen battery.
3. Explain what will happen to the concentration of the reactants as the battery is used.
4. Explain what you think has happened when a battery goes flat.

Extension

Some types of batteries are rechargeable. Find out what happens to the concentration of the reactants when these batteries are recharged.

What did you discover?

Explain how our understanding of the rates of reactions helps us understand how batteries work.

The recharging of batteries is associated with reversible reactions. You will learn about reversible reactions in Units 3 & 4.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)

The Haber process – reactions in industry

Early in the 20th century, a shortage of naturally occurring nitrogen fertilisers led to the development of a method of producing ammonia. A German chemist, Fritz Haber, refined the conditions of the reaction to maximise the yield of ammonia in satisfactory conditions. This is known as the **Haber process**. One major factor that he needed to consider was the reaction rate. If the rate was too slow, not enough ammonia would be produced in a given time.

During the Haber process, nitrogen gas is combined with hydrogen gas in a 3:1 ratio. These reactants undergo the following exothermic reaction.

$$3H_2(g) + N_2(g) \rightarrow 2NH_3(g)\ \Delta H = -92.4\ \text{kJ mol}^{-1}$$

This can be represented by an energy profile diagram, as seen in Figure 4.10.

THE HABER PROCESS

View this animation to understand how the Haber process occurs.

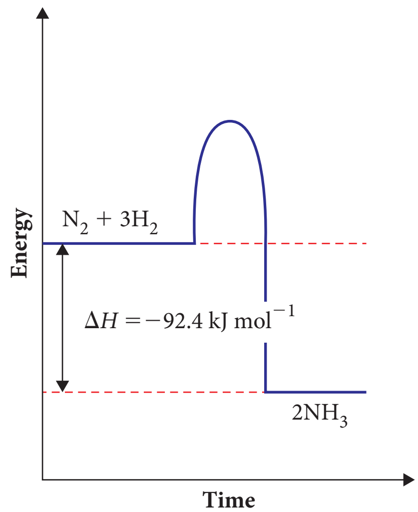


Figure 4.10 Energy profile diagram for the Haber process

Fastest production of ammonia

You have previously learnt about the factors that may affect the rate of a reaction. We can use this knowledge to understand the conditions that will allow the fastest production of ammonia in the Haber process.

ACTIVITY 4.6

RATE OF THE HABER PROCESS

Aim

To use your understanding of collision theory and the factors that affect the rate of a reaction to identify the conditions that would produce ammonia at the fastest rate

You will need

- Computer with Internet access

What to do

1 Explain how temperature could be used to increase the rate of the reaction for the Haber process. What factors do you think may limit the temperature that could be used?
2 Explain how the surface area is increased by the reactants being gases. How does this influence the rate of the reaction?
3 Explain how the pressure of the system will affect the concentration of the reactants. Would a high or low pressure increase the rate of the production of ammonia? What may limit the extent to which pressure can be manipulated?
4 Research which catalysts are used in the Haber process and how they are supplied in the system. How are the catalysts used to ensure they are the most efficient?

What did you discover?

Present your findings in a concept map to clearly demonstrate your understanding of how each factor can be manipulated to increase the rate of production of ammonia in the Haber process. For each, use collision theory to explain how each factor is able to affect the rate of the reaction.

Conditions used by industry

For the industrial production of ammonia using the Haber process, a temperature of 400–450°C and pressure of 200 atm (20 Mpa) is used. Yet these conditions do not produce the maximum rate of reaction. Why would the manufacturers use conditions that produce ammonia at a slower rate than what is possible?

There are many factors to consider when deciding on the conditions for a reaction. One is the **yield** of the reaction. This means how much of the reactants become products.

Another factor to consider is the cost. It is very expensive to have very high temperatures or pressure. There is a point where the revenue from the extra ammonia produced is outweighed by the cost of maintaining the high temperature and pressure.

When considering the conditions for a reaction, both the rate of the reaction and the yield of the reaction need to be considered. The rate of the reaction is discussed in Units 1 & 2, while the yield will be covered during your study of equilibrium in Units 3 & 4.

To learn about quantities of gases and how to calculate these, refer to Chemistry sections 8.4 and 8.5 on pages 355 and 361.

CHAPTER SUMMARY

- Collision theory allows us to explain the requirements of a reaction and understand factors affecting the rate of a reaction. We are able to alter the rate of reactions by changing the concentration or pressure of reactants, temperature, surface area or presence of catalysts.
- Organic catalysts are known as enzymes. They are usually made of proteins and are specific for a certain reaction. Enzymes are affected by concentration, coenzymes, temperature and pH.
- Cellular respiration is a chemical reaction that occurs in the mitochondria of cells to produce energy. The rate of cellular respiration is dependent on the concentration of oxygen and glucose.
- Digestion provides the body with the required nutrients for survival. Chewing food helps break the food up into smaller pieces, thereby increasing the surface area available for chemical reactions. This increases the rate of digestion.
- The normal human body temperature is approximately 37°C. If the temperature falls too low, then the rate of the reactions in the body decreases to a dangerous level.
- Fires are an example of a chemical reaction. The rate of the reaction for a fire can be manipulated by chopping the wood into smaller pieces or controlling the amount of oxygen available.
- Food spoilage is the breakdown of the food, causing it to change its taste, texture or smell. Food spoilage occurs when micro-organisms or enzymes cause a chemical reaction that degrades the food. Food preservation techniques utilise different methods of killing micro-organisms, slowing the growth of micro-organisms or slowing the chemical reaction.
- Batteries provide electrical energy through a chemical reaction. When the concentration of reactants is low, the rate of the chemical reaction is not fast enough to produce sufficient electricity and the battery is 'flat'.
- The Haber process is an industrial process that produces ammonia from nitrogen and hydrogen gases. The Haber process uses various techniques to increase the rate of the reaction to produce ammonia, including increasing the temperature, increasing the pressure and using a catalyst. However, other factors must also be considered in the Haber process such as cost, safety and yield.

CHAPTER GLOSSARY

active site the section of an enzyme where the substrate(s) binds and undergoes a chemical reaction

amino acid a small organic molecule that combines to form proteins

biochemistry the study of chemicals and their reactions in living organisms

carbohydrate an organic molecule made up of one to many molecules of simple sugars

cellular respiration the reactions that occur in cells to convert the energy in nutrients into energy that can be used; many organisms rely on glucose and oxygen to react to produce carbon dioxide, water and energy

co-enzyme a non-protein chemical that binds to an enzyme and is necessary for the function of the enzyme

collision theory a theory that explains the rate of reaction at a molecular level; it states that, for a reaction, the particles must collide with sufficient energy and the correct orientation

combustion a reaction with oxygen to form the oxides of each of the elements present; with adequate oxygen, combustion of a hydrocarbon will produce carbon dioxide and water

denature altering the chemical structure so that the original properties are lost

enzyme a protein molecule that catalyses a specific type of reaction by lowering the energy of activation

fat an organic molecule that is insoluble in water; many are made up of fatty acids joined to glycerol

food preservation methods of slowing the degradation or spoiling of food

food spoilage negative changes in food causing changes in taste, smell and feel

Haber process an industrial process of ammonia production from nitrogen and hydrogen gases

homeostasis processes that maintain the stable internal environment of an organism within limits

hypothermia the condition of a reduced body temperature (below 35°C), lower than needed for normal metabolism

kinetic energy the energy of movement

micro-organism a very small organism that can be seen only with the use of a microscope such as a bacterium, virus, or protozoan

protein a large organic molecule made up of many amino acids joined together

proteolytic enzyme an enzyme that breaks proteins into smaller lengths

rate how much one quantity changes with respect to another quantity

substrate the chemical that fits into the active site of an enzyme

yield the amount of product produced during a reaction